

Multi-Channel Integration

Part 3: Meta Webhooks & Messenger Gateway

researchcontentlab@gmail.com

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Webhook Architecture Flow

- FastAPI webhook routers handle inbound DMs from Facebook and Instagram.
- Path: `api/routes/facebook.py` & `api/routes/instagram.py`
- High-level flow:
 1. Facebook user sends a message.
 2. Meta forwards payload to FastAPI webhook.
 3. Webhook parses text and executes `engine.run_text()`.
 4. Webhook posts response back to Meta Graph API.

Active Subscriptions:

- `messages`
- `messaging_postbacks`
- `message_reads`
- `message_deliveries`

Secure Webhook Signature Validation

- Webhook endpoints verify Meta signatures (x-hub-signature-256) to block fraudulent payloads.
- Computes SHA256 HMAC of the request body using META_APP_SECRET.

```
body = await request.body()
signature = request.headers.get("x-hub-signature-256", "")
expected = "sha256=" + hmac.new(
    META_APP_SECRET.encode(), body, hashlib.sha256
).hexdigest()

if not hmac.compare_digest(signature, expected):
    return PlainTextResponse("Forbidden", status_code=403)
```

Meta Messaging API Client

- Core module: `api/routes/facebook.py`
- Sends responses back to Messenger asynchronously using the **aiohttp** HTTP client.

API Send Code:

```
async def send_messenger_message(recipient_id: str, text: str):
    url = f"https://graph.facebook.com/v20.0/me/messages?access_token={PAGE_TOKEN}"
    payload = {
        "recipient": {"id": recipient_id},
        "message": {"text": text}
    }
    async with aiohttp.ClientSession() as session:
        async with session.post(url, json=payload) as resp:
            return resp.status == 200
```